

## Transformations of $f(x) = |x|$ , Part I - Practice

Algebra II | Unit 2 | Lesson 2 Practice | TEK A2.6.C

**Objective:** I can describe the effect of  $a$  and  $b$  on  $f(x) = a|bx|$ , including stretches, compressions, and reflections.

### Section 1 — Recall

1. Describe the transformation of  $f(x) = 6|x|$  from the parent function.

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2. Describe the transformation of  $f(x) = (1/4)|x|$  from the parent function.

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### Section 2 — Apply

3. Describe the transformation of  $f(x) = -2|x|$  from the parent function.

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4. Which graph is flatter:  $f(x) = (1/3)|x|$  or  $f(x) = 5|x|$ ? Explain using the  $a$ -values.

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### Section 3 — Challenge

5. Functions  $f(x) = 7|x|$  and  $g(x) = |bx|$  produce the SAME graph. What value of  $b$  makes this true?

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6. Order from flattest to steepest:  $|x|$ ,  $-4|x|$ ,  $(1/2)|x|$ ,  $2|x|$

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### Section 4 — Explain It

7. Explain why  $f(x) = |-x|$  always produces a graph identical to the parent function  $f(x) = |x|$ , no matter what.

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